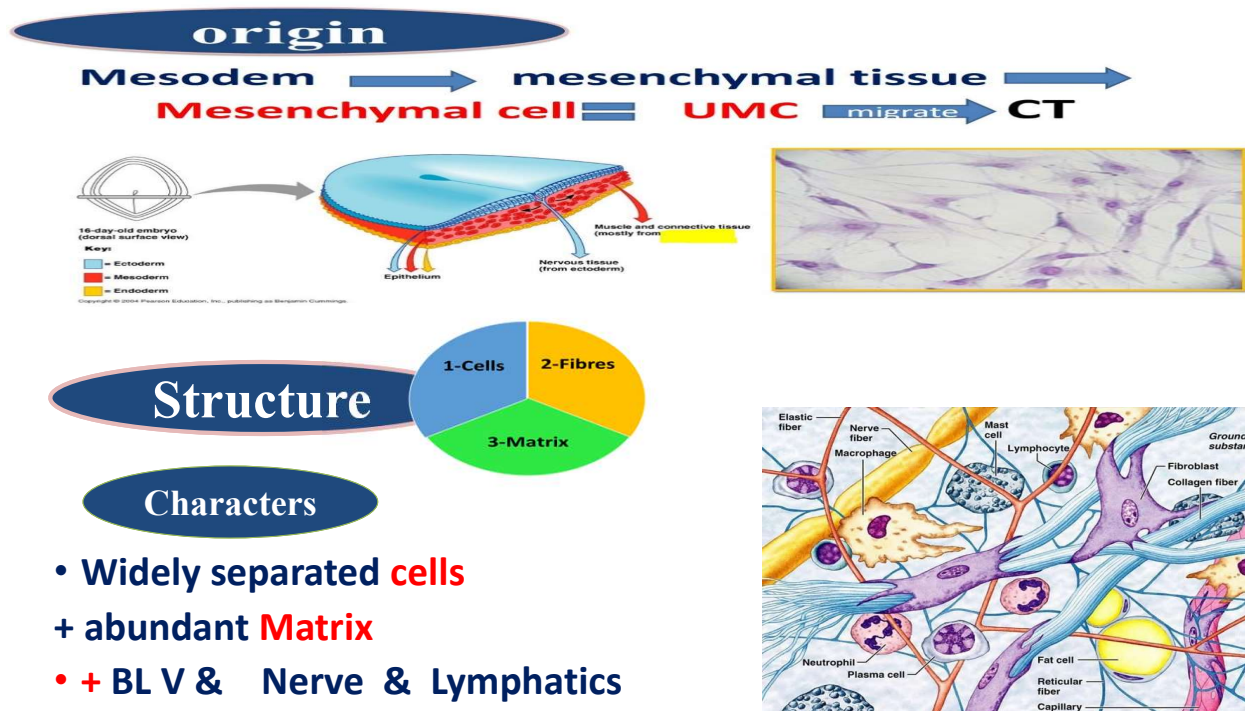


بسم الله الرحمن الرحيم

CONNECTIVE TISSUE

By
Prof . Eman Abas



Types of CT

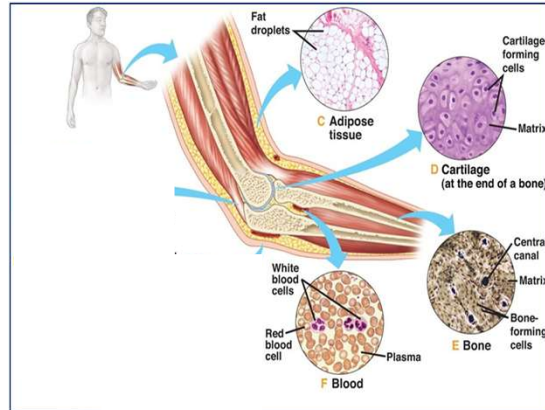
Matrix???

Soft =CT proper

Firm =Cartilage

Solid =Bone

Fluid =Blood



Structure of CT proper

Cells

1. Origin
2. Site
3. LM (shape+ Cytop+ N +stain)
4. EM
5. FN

Fibres

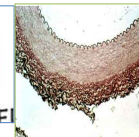
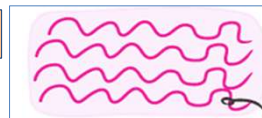
White Collagen

Strength



Yellow elastic

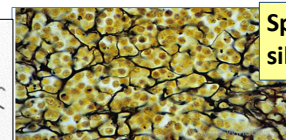
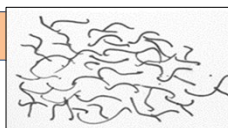
Elasticity



Special stain:
Orcein : brown

Reticular

Support



Special stain:
silver : black

= 3s

Matrix

Soft

Cells

1-Fibroblast

- ❖ **Origin:** UMCs
- ❖ **Site:** **commonest** cell/ in all ct proper
- ❖ **LM:**
 - Branched with **many long** processes
 - N: oval- **Pale** basophilic with prominent nucleolus
 - Cytop: **Deeply** basophilic
- ❖ **EM:** Features of **protein forming cells=**
- ❖ **Functions:**
 - **Synthesis** of CT **fibers**
 - **Synthesis** of **matrix**
 - Wound Healing & repair

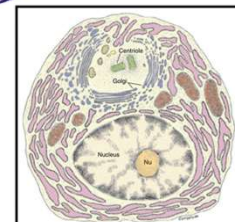
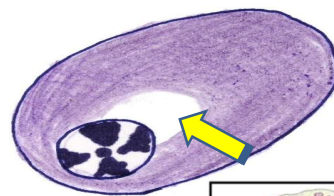
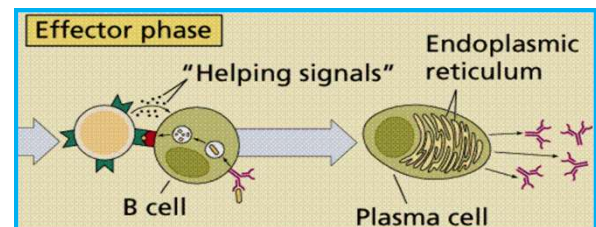
By LM:
1- N: PBP
2- cytopl: Deep basophilic

By EM:
1- Euchromatic nucleus
2- Many rER
3-Many mitochondria
4-Well developed GA



2-Plasma cells

- ❖ **Origin:**
 - B- LYMPHOCYTE → plasma cell
- ❖ **Site:** ++ lymphatic tissue
- ❖ **LM:**
 - **oval**
 - Cytoplasm: Deep **basophilic** with **-ve** Golgi image
 - N: eccentric rounded with
 - **clock-face** or **cart wheel** appearance
- ❖ **EM:**
 - protein forming cells=.
- ❖ **Function:** Antibodies formation & secretion



❖ **Origin:** UMCs

❖ **Site:**

- Ct around BV.
- Ct under skin
- CT in Resp
- CT in GIT

❖ **LM:**

- Large **Oval** cells
- Nuclei: **central**, rounded, pale
- Cytop: **basophilic granules**

(metachromatic stained with

Toluidine blue → purple or red

❖ **EM:** Features of Ptn forming cells

+

Electron-dense granules

By EM:

- 1- Euchromatic nucleus
- 2- Many rER
- 3- Many mitochondria
- 4- Well developed GA

function

Histamine:

VD& start allergy

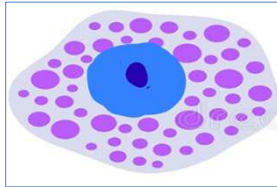
Heparin:

anticoagulant

ECF

Attract eosinophil which release histaminase and heparinase and terminate allergy

3-Mast cells



4-Unilocular Fat cell=

❖ **Site:** White adipose CT

❖ **LM :** Large **oval**

- **Single** large **fat droplet**
- small rim of cytop.
- **N:** flat & peripheral

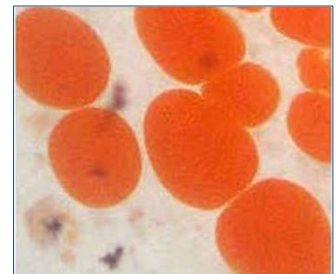
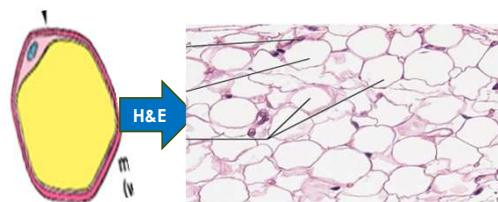
○ **H&E:** → **Signet ring appearance**

○ **Sudan III** → **orange**

○ **EM:** large single fat droplet

❖ **Functions:**

- Fat **synthesis & storage**
- Heat insulation
- Support organs



4 IMPORTANT TYPES OF CT PROPER

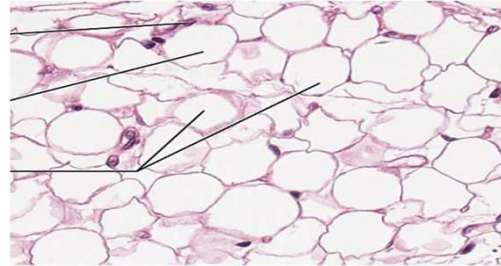
1-Adipose CT

Structure:

Cells: mainly fat cells

Site

Widely distributed:
under skin
Mammary gland
Around kidney



Fn:

Store fat

Heat Insulator

Shape= body contour

Support Kidney

2-Reticular CT

3S

❖ **Structure:**

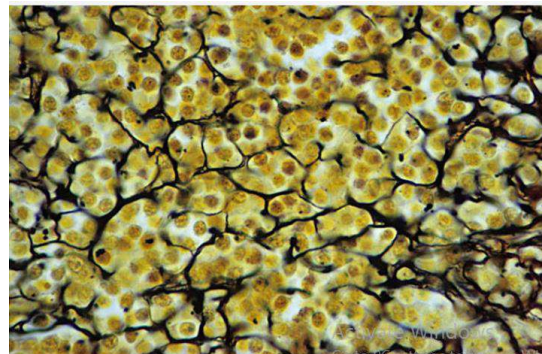
□ Reticular cell =

□ Reticular fibres

○ **Stain** Silver: → br-black

❖ **Site&Fn:** stroma

□ Support



3-white fibrous CT

Structure:

Fibres: +++collagen
bundles

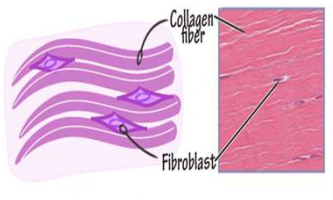
Cells: fibroblasts

MX: Minimal

2 Types:

Regular

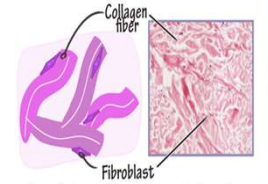
Regularly
arranged collagen
bundles
with **rows** of FB



function : Withstand=RESIST stretch(IN 1 direction)
Site : Tendon

Irregular

Irregularly
arranged collagen
bundles
with FB



function : Withstand stretch
(IN different directions)

Site : skin

4-Yellow elastic CT

Structure:

Fibres: +++EF

Cells: fibroblasts

MX: small amount

- Fresh state → yellow
- Staining: **Orcein:** → brown

Function:

Elasticity= recoil when stretched

sites

**Aorta &
large Arteries**

**Trachea
bronchi
bronchioles**

